

PRANAY THARALA

Artificial Intelligence & Machine Learning Engineer

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PROFESSIONAL SUMMARY

- AI/ML Engineer with 7+ years of experience building and deploying production machine learning and generative AI systems across financial services, driving measurable gains in accuracy, automation, and regulatory compliance.
- Expert in LLM fine-tuning (LoRA/QLoRA on Llama, Mistral) and Retrieval-Augmented Generation (RAG) pipelines using LangChain, Pinecone, FAISS, OpenSearch, and pgvector, delivering context-aware, cited answers over enterprise fund, account, and transaction documents.
- Proficient in Python, Java, JavaScript, AWS (SageMaker, Bedrock, Lambda), GCP Vertex AI, and Azure AI, applying MLOps/LLMOps practices (MLflow, Kubeflow, Airflow, DVC) for end-to-end model deployment, monitoring, and retraining.
- Built and deployed machine learning models (XGBoost, Random Forest, Gradient Boosting) and NLP pipelines (BERT, spaCy, Hugging Face Transformers) for fraud detection, risk scoring, and document intelligence, reducing false positives and manual review time.
- Hands-on experience with optical character recognition (OCR) and document-intelligence solutions that extract and process data from documents, forms, images, and PDFs, and with AI-driven decision-support tools for high-risk transaction reviews and customer risk assessment.
- Engineered cloud-native APIs and microservices (Java, Spring Boot, AWS Lambda) bridging AI platforms with enterprise data systems, supporting responsible-AI reviews and SEC/FINRA-adjacent regulatory requirements.
- Effective communicator and mentor, partnering with engineers, product managers, and business stakeholders to translate requirements into production-grade AI solutions, and guiding junior engineers on AI architecture and CI/CD best practices.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, R, Linux

Frameworks & Libraries: Spring Boot, Angular, TensorFlow, PyTorch, scikit-learn, Hugging Face, XGBoost, LangChain, LangSmith, FastAPI, Flask, BERTopic, RetinaNet, Faster R-CNN, BERT, Transformers

Databases & NLP: SQL, NoSQL, Oracle, PostgreSQL, MySQL, MongoDB, Cassandra, BigQuery, Vector Databases (Pinecone, FAISS, OpenSearch, pgvector, Chroma), Graph Databases (Neo4j), Natural Language Processing (NLP), Agentic AI Frameworks

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering (K-Means, Hierarchical), Decision Trees, Random Forest, Gradient Boosting (XGBoost, LightGBM), SVM, KNN, Naive Bayes, Ensemble Methods, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Regularization (L1, L2), Dimensionality Reduction (PCA, t-SNE), Time Series Forecasting (ARIMA, SARIMA, Prophet, LSTM), Model Interpretability (SHAP, LIME), Recommendation Systems

Deep Learning: ANN, CNN, RNN, LSTM, Bi-LSTM, GRU, Autoencoders, Transfer Learning (VGG, ResNet, InceptionNet), Attention Mechanism, Object Detection (YOLO, Faster R-CNN, RetinaNet), Image Segmentation (U-Net, Mask R-CNN), OCR, Optimization Algorithms (SGD, Adam, RMSprop), Model Deployment

NLP Techniques: Tokenization, Lemmatization, Stemming, TF-IDF, Word Embeddings (Word2Vec, GloVe, FastText), Seq2Seq Models, Transformer Models (BERT, GPT, T5), NER, Sentiment Analysis, Text Classification, Text Summarization, Question Answering, Topic Modeling (LDA), spaCy, NLTK

Generative AI: LLMs, Ollama, Gemini (familiar), LangChain, LangSmith, MCP, Agentic AI, Fine-Tuning (LoRA, QLoRA), Local LLMs (Mistral, Llama, Gemma), Inference APIs (Groq, Nvidia NIM), Retrieval-Augmented Generation (RAG), Prompt Engineering, Prompt Templates, Document Chunking & Indexing, GANs, Diffusion Models, Text-to-Image Generation (DALL-E, CLIP)

MLOps & LLMOps: CI/CD for ML, Model Deployment (REST API, Batch), Model Monitoring, Model Versioning, Retraining Pipelines, Experiment Tracking (MLflow, Weights & Biases), Data Versioning (DVC), Feature Stores (Feast), Kubeflow, LLM Response Evaluation (TruLens, PromptLayer), Scalable LLM Deployment (TGI, vLLM)

Big Data: Hadoop, Hive, HBase, Apache Spark, PySpark, Kafka, Pub/Sub, Apache Beam, Sqoop

Tools & Software: Excel, Tableau, Power BI, Pandas, NumPy, Docker, Kubernetes, Terraform, Git, Jenkins, GitLab CI, GitHub Actions, Cloud Build, Grafana, Splunk, Prometheus, MLflow, Airflow, Agile, VS Code, Jupyter Notebook, Google Colab, Postman

Cloud Platforms: AWS (SageMaker, Bedrock, EC2, S3, RDS, Lambda, CloudFormation), GCP (Vertex AI, BigQuery, Dataflow, Cloud Run), Azure (Functions, Azure AI, Azure ML, Azure OpenAI)

Soft Skills: Requirements gathering, Cross-functional collaboration, Mentoring, Stakeholder communication, Business reporting

WORK EXPERIENCE

American Express | Mar 2026 - Present

Senior AI Engineer

- Ramped up on American Express's enterprise AI platform, internal LLM tooling, data-governance standards, and prompt-engineering guidelines within the first weeks, accelerating readiness to build and deploy ML and generative AI solutions for financial-services use cases.
- Partner with engineers and product managers to scope and deliver production-grade RAG and agentic AI virtual-assistant initiatives, supporting fraud-detection and document-intelligence workstreams through end-to-end model deployment.
- Design and integrate LLM-powered agents and event-driven APIs into enterprise applications, leveraging LangChain and vector databases (Pinecone, FAISS) to power retrieval-augmented NLP and agentic workflows.
- Build AI-driven decision-support tools that analyze high-risk transactions and assist analysts with fraud-review and customer risk-assessment decisions.
- Evaluate and benchmark LLM providers and vector-database options for upcoming production pipelines by running systematic latency, cost, and accuracy trade-off experiments, informing platform-wide build vs. buy decisions.
- Support responsible-AI reviews and observability checkpoints by documenting model-evaluation criteria and bias-mitigation controls throughout training and deployment cycles, strengthening audit readiness.
- Apply machine learning models (XGBoost, Gradient Boosting) alongside rule-based logic to automate anomaly detection in fraud workflows, reducing false positives and manual investigation effort.
- Apply NLP techniques (BERT, Hugging Face Transformers, Named Entity Recognition) and optical character recognition (OCR) to document-intelligence workflows, extracting and classifying entities across compliance and customer documents to speed review cycles.
- Use Claude daily to build, evaluate, and operate production agentic AI systems and RAG pipelines, accelerating delivery of AI features.

Environment: AWS, GCP Vertex AI, LangChain, Pinecone, FAISS, LLM Agents, Claude, Prompt Engineering, RAG, XGBoost, BERT, Hugging Face, NLP, Python, Financial Services

Vanguard | Sep 2023 - Mar 2026

AI/ML Engineer

- Architected and deployed AI engineering pipelines that generated 2-page automated summaries of unstructured fund disclosures, account statements, and investment-operations documents by integrating AWS Lambda, Terraform, and REST APIs, cutting manual document-review effort for operations teams.
- Designed prompt-engineering strategies — including hybrid prompting techniques — and fine-tuning (LoRA/QLoRA on Llama, Mistral) for domain-specific generative AI solutions, reducing hallucinations and improving compliance accuracy across investment and brokerage workflows.
- Led and mentored a cross-functional team on an AI/ML modernization roadmap, translating legacy fund- and account-administration challenges into scalable AI solutions adopted across the platform.
- Deployed RAG pipelines (LangChain, Pinecone, FAISS, Vertex AI) over legacy fund and account documentation, evaluating OpenSearch and pgvector as retrieval options, enabling modern API-first access to historical records.
- Engineered cloud-native APIs and microservices (Java, Spring Boot, AWS Lambda) and event-driven integrations bridging AI platforms with enterprise data systems, improving cross-system interoperability.
- Evaluated and selected LLM providers (OpenAI API, AWS Bedrock, GCP Vertex AI) for production use by weighing latency, cost, and accuracy against legacy system constraints through systematic experimentation.
- Contributed to responsible-AI reviews, model evaluation, and bias-mitigation checkpoints, ensuring AI solutions met financial-services and SEC/FINRA-adjacent regulatory requirements during training and deployment.
- Developed and deployed ML models (XGBoost, Random Forest, Gradient Boosting) for customer risk assessment, risk scoring, and anomaly detection, improving early-detection accuracy and reducing false positives in transaction and account activity.
- Built NLP pipelines (BERT, spaCy, Hugging Face Transformers) to extract entities and classify unstructured investment and account documents, improving processing accuracy and reducing manual review time.
- Implemented time-series forecasting models (ARIMA, Prophet) for portfolio and operational metrics, improving prediction accuracy by up to 25%.
- Applied MLOps practices (MLflow, Airflow) for experiment tracking, model versioning, and automated retraining pipelines, ensuring reliable, production-grade ML systems.
- Mentored junior engineers on AI architecture, prompt-engineering best practices, and CI/CD deployment patterns for production machine learning systems.

Environment: AWS (Lambda, Bedrock), GCP Vertex AI, LangChain, Pinecone, FAISS, OpenSearch, pgvector, Terraform, Java, Spring Boot, LoRA/QLoRA, Llama, Mistral, OpenAI API, XGBoost, BERT, spaCy, Hugging Face, ARIMA, Prophet, MLflow, Airflow

HCLTech | Jul 2020 - Jul 2023

Software Engineer

- Promoted to full-time Software Engineer after completing the HCL TechBee apprenticeship, taking ownership of feature development, testing, and deployment for global client applications.
- Designed, developed, and maintained backend services and RESTful APIs using Java and Spring Boot, following object-oriented design principles and enterprise coding standards.

- Built reusable, responsive front-end UI components using Angular (RxJS, Reactive Forms) and JavaScript, and integrated them with secure backend microservices, improving application performance and consistency across client-facing portals.
- Deployed and supported cloud infrastructure on AWS (EC2, S3, RDS, Lambda) and Microsoft Azure, automating infrastructure provisioning with Terraform for consistent, repeatable deployments.
- Wrote and maintained unit and integration test suites (JUnit, Mockito), improving code coverage and reducing regression defects across release cycles.
- Participated in Agile/Scrum ceremonies with distributed, cross-functional teams to deliver features on schedule.
- Performed peer code reviews and enforced version-control best practices (Git, branching strategies, pull requests) to maintain code quality across the team.
- Built and maintained CI/CD pipelines (Jenkins, GitLab CI, Docker, Kubernetes) to automate build, test, and deployment steps, shortening release cycles, enabling zero-downtime rollouts, and cutting manual deployment effort by over 30%.
- Wrote and optimized SQL queries across Oracle, PostgreSQL, and MySQL, applying indexing and caching strategies to reduce latency, and supported schema changes and data migrations for relational databases.
- Provided L2/L3 production support, triaging incidents, performing root-cause analysis, and coordinating fixes to meet SLA commitments and minimize downtime.
- Implemented monitoring and observability practices using Grafana and Splunk to track application health, accelerate incident detection, and improve production stability.
- Interfaced directly with client stakeholders to gather requirements, clarify specifications, and communicate technical trade-offs and delivery timelines.
- Authored technical documentation (design notes, runbooks, release notes) to support knowledge transfer and long-term maintainability.

Environment: Java, Spring Boot, Angular, JavaScript, AWS, Azure, Terraform, JUnit, Mockito, Git, Jenkins, GitLab CI, Docker, Kubernetes, Oracle, PostgreSQL, MySQL, Grafana, Splunk, SQL, Agile/Scrum, REST APIs

HCLTech | Jul 2019 - Jul 2020

Technology Trainee - HCL TechBee (Apprenticeship)

- Selected from a highly competitive nationwide applicant pool for a 12-month work-integrated apprenticeship combining structured technical training with on-the-job experience.
- Completed a structured curriculum covering programming fundamentals (Java, Python), data structures and algorithms, object-oriented programming, RDBMS/SQL, and the software development lifecycle (SDLC).
- Built and presented mini-projects applying classroom concepts to practical coding exercises, reinforcing fundamentals in debugging, testing, and code readability.
- Rotated onto live client engagements under senior engineer supervision, gaining hands-on exposure to enterprise version control (Git), issue tracking (JIRA), and production deployment workflows.
- Practiced technical communication, ticket documentation, and collaborative problem-solving within an Agile team environment.
- Completed periodic technical assessments and a capstone evaluation, earning conversion to a full-time Software Engineer role.

Environment: Java, Python, SQL, Git, JIRA, Agile, SDLC

PROJECTS

Enterprise Document-Intelligence RAG Assistant: Built a retrieval-augmented Q&A assistant using NLP over fund and account documents (LangChain, Pinecone, FAISS, GCP Vertex AI), surfacing cited, context-aware answers that cut lookup time for compliance and support teams.

Fine-Tuned Financial-Domain LLM: Applied LoRA/QLoRA fine-tuning to open-source LLMs (Llama, Mistral) through iterative training on compliance and transaction datasets, reducing hallucinated responses and improving domain accuracy.

Fraud-Detection Virtual Assistant: Designed an AI-powered decision-support virtual assistant integrated with vector databases to triage high-risk fraud alerts, cutting manual investigation time for analyst teams.

EDUCATION

- Master's Degree, Information Technology - St. Francis College
- Bachelor of Computer Applications (BCA) - SASTRA University